

Do Green Industrial Policies Transform Sustainability into Efficient Productivity?

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The intensifying climate crisis and growing ecological constraints have fundamentally altered the way governments design industrial policy. Environmental sustainability is no longer viewed as a peripheral concern but as a central pillar of long-term economic development. In this context, **Green Industrial Policies (GIPs)** have gained prominence as policy frameworks that deliberately integrate environmental objectives with industrial and productivity goals. Unlike conventional industrial policies that prioritize output expansion and competitiveness alone, green industrial policies aim to steer structural transformation toward cleaner, more resource-efficient production systems. A critical question emerging from this shift is whether such policies merely impose environmental compliance costs or whether they actively transform sustainability into a source of efficient productivity. This article examines this question by reviewing the theoretical foundations, empirical evidence, and firm-level responses associated with green industrial policies.



Powering productivity through green innovation

Green industrial policy refers to **government strategies** that combine climate goals with industrial growth. These are mainly **sector- targeted policies** that affect the economic production with the aim of generating environmental benefits. Key tools of these policies include clean technology subsidies, carbon pricing, green public procurement, and industrial decarbonisation programs. From a development perspective, these policies drive structural transformation by pushing industries toward higher productivity, optimal use of resources, and more advanced

technologies. As global markets increasingly demand sustainability compliance, such policies are becoming central to industrial competitiveness.

Environmental literature has been detrimental throughout to firm performance due to increasing compliance costs. In this context the **Porter Hypothesis** suggests that strict environmental regulation stimulates the discovery and introduction of new technologies leading to efficiency gains that offset or even exceed regulatory costs. According to **Rodrik (2014)**, such policies are most effective when they target both environmental externalities and coordination problems in the emerging green sectors. The paper by **Piserà, Gandullia, and Girardone (2025)** builds on an extensive strand of research investigates the linkage between engagement to firm level productivity and financial performance measured through total factor productivity. The productivity advantage of ESG is most apparent during times of economic downturn, indicating a resilience factor. In general, this research repositions ESG engagement as a strategic productivity force at the firm level rather than a compliance or reputation-building exercise. **Quality & Quantity, 2025** study finds that the sustainability-related uncertainty in ESG practices hinders green growth in the US and UK, while human capital development and innovation in technology are important factors that contribute to sustainable economic growth. The authors have also demonstrated that good governance and market-oriented policies are essential in overcoming ESG uncertainty and promoting green growth. Evidence from countries such as Germany, South Korea, and China indicates that green policy frameworks have helped these countries achieve productivity gains in the renewable energy and clean manufacturing sectors through technological advancements and economies of scale.

Green Industrial Policies affect firm level productivity through **3 main channels**, namely **Technology, Input efficiency** and **Research & Development**. Green industrial policies encourage adoption of green technology which often leads firms to upgrade production equipment, efficient energy usage and improving output quality. Secondly, energy and resource efficiency measures can reduce the long run cost of operations of the firms and stabilize production expenses. Green policies cause a spillover effect on innovation which in turn stimulates Research & Development, generating productivity gains across supply chains. Apart from the supply chain it also helps in horizontal and forward linkages. Recent global industry data shows that firms adopting energy-efficient technologies are achieving **5–18% short-term energy cost savings**, with **long-term savings** reaching **40–60%** and improvement in output per worker. However, productivity gains typically appear after initial adjustment periods, highlighting the importance of policy stability and long-term investment signals.

The table given below shows the Global Evidences on Green Industrial Policies, Industry Responses and Productivity Outcomes :

Region / Country	Major Green Industrial Policy	Firm / Industry Response	Evidence on Productivity / Efficiency Impact
India	Renewable auctions, PLI for solar & batteries, ESG push, industrial decarbonisation policies	Rising corporate green investment, renewable adoption, green supply chain transition	Corporate environmental CSR spending in industrial Gujarat rose ~90% in FY2023-24, showing rapid shift toward sustainability-linked competitiveness
USA	Inflation Reduction Act (IRA), clean manufacturing subsidies, EV incentives	Large-scale domestic clean manufacturing investment	Clean tech investment expanding production capacity and lowering long-run energy cost exposure
European Union	EU Green Deal, Carbon pricing (ETS), circular economy regulation	Energy efficiency upgrades, low-carbon process innovation	OECD studies suggest environmental regulation can raise green productivity when policy stringency is moderate
Australia	Safeguard Mechanism, hydrogen and renewable industrial transition programs	Emissions monitoring + cleaner fuel switching	Gradual efficiency gains via process optimisation and energy cost stabilisation
OECD (Cross-country evidence)	Environmental policy stringency frameworks	Innovation adoption and technology upgrading	Evidence supports Porter Hypothesis under moderate regulation — productivity rises via innovation offset

The response of firm-level measures to green industrial policies is diverse. Large firms, which have better access to finance and technology, tend to respond positively to green policies as an opportunity for innovation and leadership in the market. Small and medium enterprises may experience some costs of adjustment in the short term, but with the help of supportive policy tools such as green credit, they can achieve efficiency gains in the long term. Notably, the quality of institutions and policy consistency has a major impact on the outcome. Research emphasizes that flexible policies with incentives tend to produce better productivity effects than rigid command policies. If policies are unstable or misdirected, productivity gains could be lagged or uneven.

Despite their potential, green industrial policies are not without challenges. The main challenges include managing **geopolitical risks** in international supply chains, closing the **financial gap** for the **decarbonization** of industry, and mitigating the risk of a “**two-speed**” **transition** that would harm developing countries. High capital costs and, in some instances, the absence of dedicated financial support for industrial sectors, which emit 23% of global CO₂ emissions but have been receiving less than proportional green finance, are also major challenges. Developing countries are challenged by their dependence on fossil fuels, while others are challenged by the lack of infrastructure for new green technologies.

Green industrial policies are increasingly shaping the future of global production. Emerging empirical evidence suggests that when these policies are well structured, supported by financial incentives and technological infrastructure, they transform sustainability to a catalyst for industrial efficiency and productivity growth. For industries, the transition towards sustainable productivity is more about long-term competitiveness rather than mere regulatory compliance. However, the long term success ultimately depends on the extent to which governments can effectively integrate environmental targets with broader development strategies.